

Modern techniques in quantum algorithms

QTM 2025 Tutorial

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The University of Tokyo
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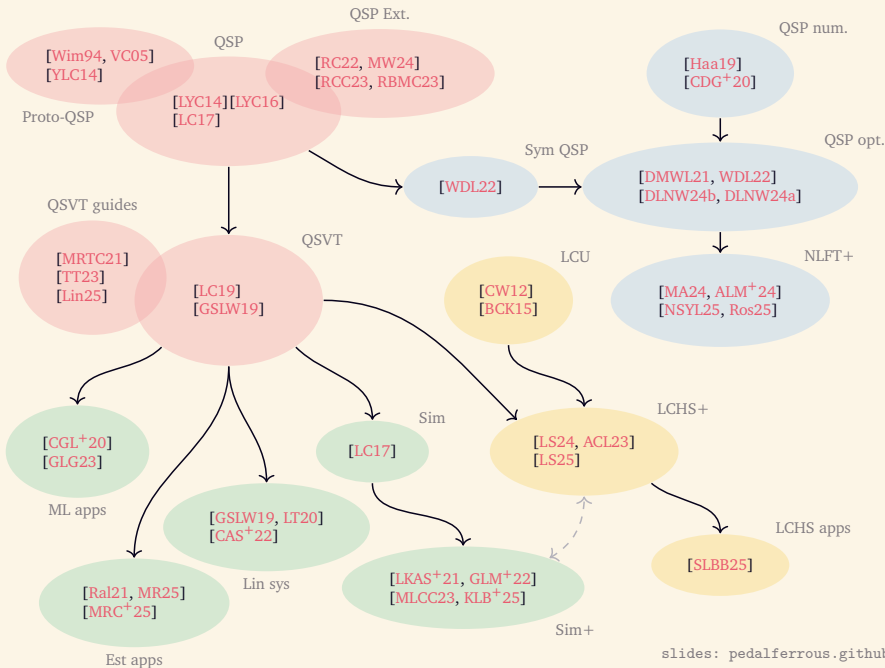


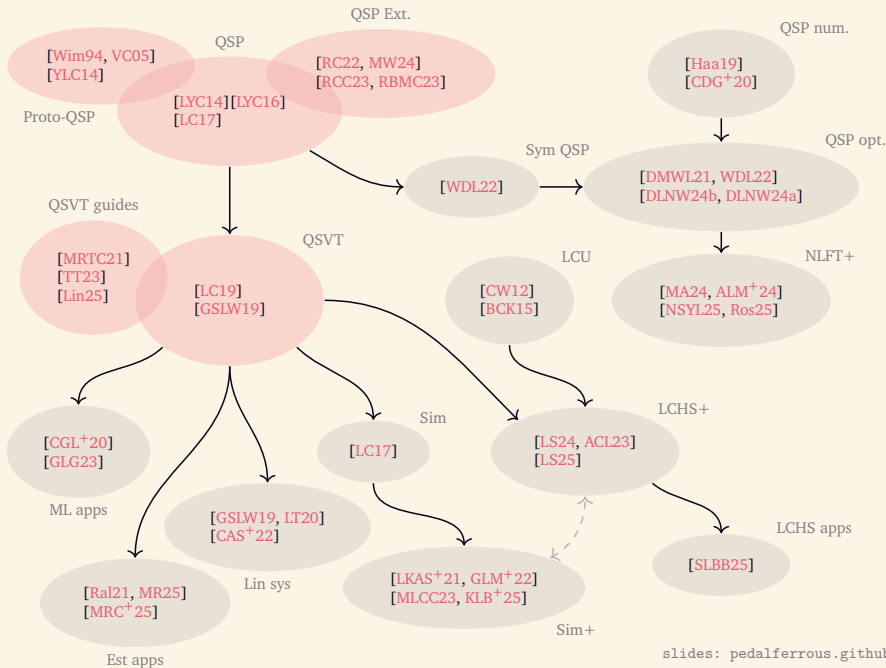
🎯 Today's aims

1. Motivation and crash-course for QSP¹ ★
2. Numerical interlude ★
3. Qubitization, block encodings, and QSVT² ★
4. Applications and the state of the art ★
5. The limits of QSVT, and alternative approaches ★
6. Open problems, recent progress, and outlook ★★☆☆

¹Quantum signal processing

²Quantum singular value transformation







NMR and composite pulses



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What can we do with many copies of an *unknown but consistent* unitary process?¹ [Wim94, VC05, BHC04]

¹E.g., Larmor precession in inhomogeneous magnetic field



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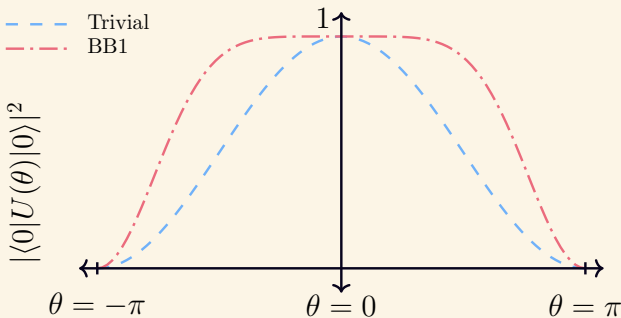
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$$\Phi \in \mathbb{R}^{n+1} \mapsto U_{\Phi}(\mathbf{x}).$$

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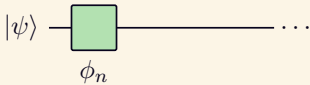
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$$W(\mathbf{x}) = \begin{bmatrix} \mathbf{x} & i\sqrt{1-\mathbf{x}^2} \\ i\sqrt{1-\mathbf{x}^2} & \mathbf{x} \end{bmatrix}, \quad e^{i\phi} = \begin{bmatrix} e^{i\phi} & \\ & e^{-i\phi} \end{bmatrix}.$$

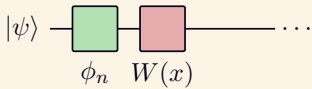
Properties of the QSP ansatz

$$|\psi\rangle \text{ ————— } \dots$$

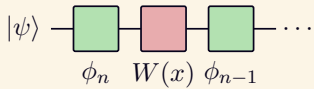
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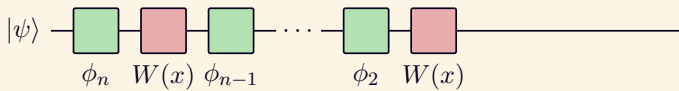
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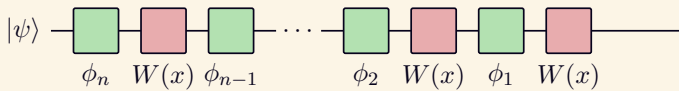
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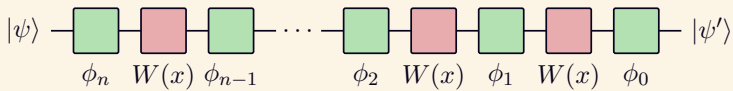
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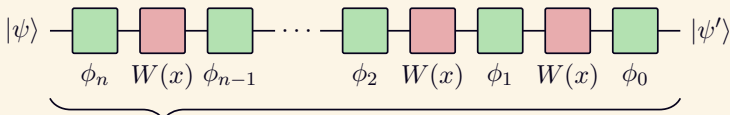
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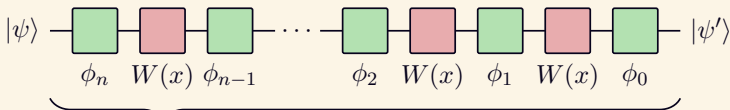
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$$U(\Phi, x) = e^{i\phi_0\sigma_z} \prod_{j=1}^k W(x) e^{i\phi_j\sigma_z} = \begin{bmatrix} P(x) & iQ(x)\sqrt{1-x^2} \\ iQ(x)^*\sqrt{1-x^2} & P(x)^* \end{bmatrix}$$

Can map $P \leftrightarrow \Phi$ efficiently; like classical filter!

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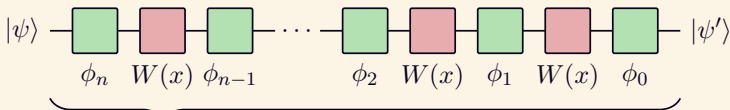


$$U(\Phi, H) = e^{i\phi_0 Z} \prod_{j=1}^k W(H) e^{i\phi_j Z} = \begin{bmatrix} P(H) & iQ(H)\sqrt{1-H^2} \\ iQ(H)^*\sqrt{1-H^2} & P(H)^* \end{bmatrix} \begin{matrix} |0\rangle \\ |1\rangle \end{matrix}$$

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Later: the theory of *block encodings*!



The BB1 Protocol¹ as QSP

¹[Wim94]



The BB1 Protocol¹ as QSP

$$U(\theta) = e^{i\pi\sigma_z/2} V(\theta) e^{-i\eta\sigma_z} V(\theta) e^{2i\eta\sigma_z} V^2(\theta) e^{-2i\eta\sigma_z} V(\theta) e^{i\eta\sigma_z}$$

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For $\eta = (1/2) \cos^{-1}(-1/4) \approx 0.912$ this yields:

$$U(\cos^{-1} x) = \begin{bmatrix} P(x) & i\sqrt{1-x^2}Q(x) \\ * & * \end{bmatrix},$$

$$|P(x)|^2 = \frac{1}{8}(30x^2 - 45x^4 + 35x^6 - 15x^8 + 3x^{10}).$$

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For small θ , $|\langle 0 | U(\theta) | 0 \rangle|^2$ expands $1 + (5/512)\theta^6 + \dots$

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 For other parameterizations ($x \mapsto [z + z^{-1}]/2$) or ansätze, this can change!

QSP ansätze and functional constraints

$$|\textcolor{red}{P}|^2 + (1 - x^2) |\textcolor{green}{Q}|^2 = 1.$$

The circuit depth for P of degree d is d . Moreover, $x \mapsto -x$ changes signs of $\textcolor{red}{P}$, $\textcolor{green}{Q}$, and $x \in \{\pm 1, 0\}$ induce constraints

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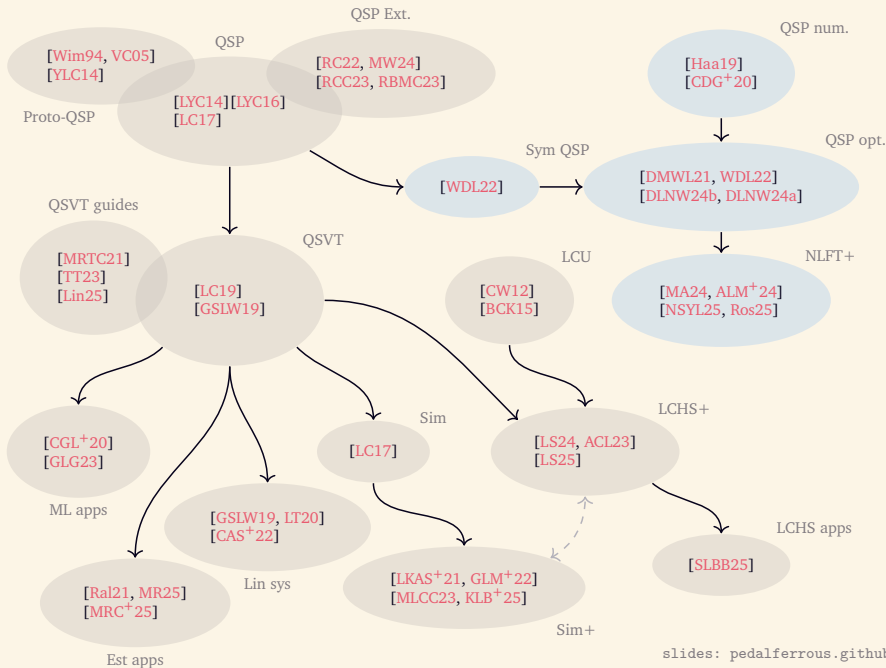
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Finding A, C reduces to *norm-constrained polynomial approximation or interpolation!*

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Extensive study of phase finding algorithms¹

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$(f, g) \mapsto (A, C)$, then² $(A, C) \mapsto (P, Q)$, then $(P, Q) \mapsto \Phi$.

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$\Phi^{t+1} = \Phi^t - [DF(\Phi^t)]^{-1}(F(\Phi^t) - A)$ quasi-Newton.

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✨ QSPPACK: <https://github.com/qspack/QSPPACK>

✨ pyQSP: <https://github.com/ichuang/pyqsp>.

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What we want to know: How can QSP *usefully* transform quantum systems comprising *many qubits*?

¹Stable, efficient classical companion algorithms



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Up next: finding qubits inside large unitaries

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Revisiting the Grover iterate



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$$\begin{aligned}|s\rangle &= \xi |t\rangle + \sqrt{1 - \xi^2} |t^\perp\rangle, \\ |t^\perp\rangle &= (1 - \xi^2)^{-1} \left[|s\rangle - \langle t|s\rangle |t\rangle \right].\end{aligned}$$

The Grover iterate alternately reflects about the target state $|t\rangle$ and the uniform superposition $|s\rangle$

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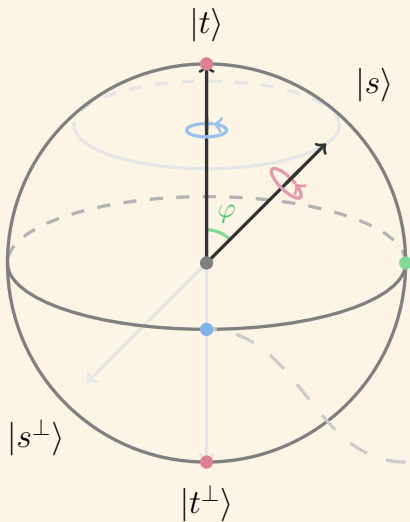
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$$-S_s(\pi)S_t(\pi) = \begin{bmatrix} -1 + 2\xi^2 & -2\xi\sqrt{1 - \xi^2} \\ -2\xi\sqrt{1 - \xi^2} & 1 - 2\xi^2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

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$$U_s(\alpha) = e^{i\alpha(2|s\rangle\langle s|-I)}$$

$$U_t(\beta) = e^{i\beta(2|t\rangle\langle t|-I)}$$

$$\frac{|t\rangle + |t^\perp\rangle}{\sqrt{2}}$$

$$\frac{|t\rangle - i|t^\perp\rangle}{\sqrt{2}}$$

$$S_s(\alpha) = I - (1 - e^{i\alpha} |s\rangle\langle s|),$$

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Let $B_{L,1} \in \mathbb{C}^{d \times r}$, $B_{R,1} \in \mathbb{C}^{d \times c}$ have *orthonormal columns*.

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A unitary $U \in \mathbb{C}^{d \times d}$ is an (α, ε) -block encoding of A if

$$\left\| A - \alpha B_{L,1}^\dagger U B_{R,1} \right\|_{\text{op}} \leq \varepsilon.$$

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E.g., a $(1,0)$ block encoding, with $B_L = (B_{L,1}, B_{L,2})$ and $B_R = (B_{R,1}, B_{R,2})$ the unitary completions of $B_{L,1}, B_{R,1}$:

$$B_L^\dagger U B_R = \begin{bmatrix} A & * \\ * & * \end{bmatrix}, \quad B_L^\dagger (\Pi_L U \Pi_R) B_R = \begin{bmatrix} A & 0 \\ 0 & 0 \end{bmatrix}.$$

An (α, a, ε) -block encoding of A [GSLW19]

$$\left\| A - \alpha(\langle 0|^{\otimes a} \otimes I)U(|0\rangle^{\otimes a} \otimes I) \right\| \leq \varepsilon,$$

where A is s -qubit, and U is $(s + a)$ qubit, and, e.g.,
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Sometimes say $\Pi_L U \Pi_R = A$, ignoring zero blocks of U .
Then A has a singular value decomposition

$$A = \sum_i \xi_i |\tilde{\psi}_i\rangle\langle\psi_i|, \quad \text{maps red to green!}$$

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Observe action of U on a **right** singular vector

$$U|\psi_i\rangle = \Pi_L U \Pi_R |\psi_i\rangle + (1 - \Pi_L) U |\psi_i\rangle \quad (1)$$

$$= \xi_i |\psi_i\rangle + \sqrt{1 - \xi_i^2} |*\rangle \quad (2)$$

Let Π_L, Π_R, U, A as before, ξ_i the i -th singular value of A , k the largest index where $\xi_k = 1$, and $r = \text{rank}(A)$.

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$d = \text{rank}(\Pi_R)$, $\tilde{d} = \text{rank}(\Pi_L)$; $|\psi_i\rangle/|\tilde{\psi}_i\rangle$ are the **right/left** singular vectors of A (bases for $\text{img}(\Pi_R)$, $\text{img}(\Pi_L)$).¹

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$$|\tilde{\psi}_i^\perp\rangle \propto (I - \Pi_L)U |\psi_i\rangle.$$

$$\mathcal{H}_i = \text{span}(|\psi_i\rangle), \quad \tilde{\mathcal{H}}_i = \text{span}(|\tilde{\psi}_i\rangle), \quad i \in [k],$$

$$\mathcal{H}_i = \text{span}(|\psi_i\rangle, |\psi_i^\perp\rangle), \quad \tilde{\mathcal{H}}_i = \text{span}(|\tilde{\psi}_i\rangle, |\tilde{\psi}_i^\perp\rangle), \quad i \in [r] \setminus [k],$$

$$\mathcal{H}_i^R = \text{span}(|\psi_i\rangle), \quad \tilde{\mathcal{H}}_i^R = \text{span}(U|\psi_i\rangle), \quad i \in [d] \setminus [r],$$

$$\mathcal{H}_i^L = \text{span}(U^\dagger|\tilde{\psi}_i\rangle), \quad \tilde{\mathcal{H}}_i^L = \text{span}(|\tilde{\psi}_i\rangle), \quad i \in [\tilde{d}] \setminus [r].$$

¹Note also $\mathcal{H}^\perp/\tilde{\mathcal{H}}^\perp$, where action of U is unspecified.

Qubitization: we have to show $\mathcal{H}_i$, $\mathcal{H}_i^R$, and $\mathcal{H}_i^L$ and so on are pairwise orthogonal:

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$$\langle \tilde{\psi}_i | \tilde{\psi}_j \rangle = \delta_{ij}, \quad \tilde{\mathcal{H}}_i \perp \tilde{\mathcal{H}}_j, \quad (4)$$

$$\langle \psi_i^\perp | \psi_j^\perp \rangle = \langle \tilde{\psi}_i^\perp | \tilde{\psi}_j^\perp \rangle = \delta_{ij}, \quad (\mathcal{H}_i)^\perp \perp (\tilde{\mathcal{H}}_j)^\perp, \quad (5)$$

$$\langle \psi_i | \psi_j^\perp \rangle = \langle \tilde{\psi}_i | \tilde{\psi}_j^\perp \rangle = 0, \quad */* \perp */*, \quad (6)$$

$$\langle \psi_i | U^\dagger | \tilde{\psi}_j \rangle = 0, \quad U^\dagger | \tilde{\psi}_j \rangle \in \mathcal{H}_j^L, \quad (7)$$

$$\langle \psi_i^\perp | U^\dagger | \tilde{\psi}_j \rangle = 0, \quad U^\dagger | \tilde{\psi}_j \rangle \in \mathcal{H}_j^L. \quad (8)$$

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The first three from singular vector orthogonality; note that $\langle \tilde{\psi}_i | U \Pi_R U^\dagger | \tilde{\psi}_j \rangle$ can be replaced by $\langle \tilde{\psi}_i | A A^\dagger | \tilde{\psi}_j \rangle$. The final three follow from definition of projectors.

The components of the QSVT circuit¹

$$U = \bigoplus_{i \in [k]} [\xi_i]_{\tilde{\mathcal{H}}_i}^{\mathcal{H}_i} \oplus \bigoplus_{i \in [r] \setminus [k]} \begin{bmatrix} \xi_i & \sqrt{1 - \xi_i^2} \\ \sqrt{1 - \xi_i^2} & \xi_i \end{bmatrix}_{\tilde{\mathcal{H}}_i}^{\mathcal{H}_i} \oplus \bigoplus_{i \in [d] \setminus [r]} [1]_{\tilde{\mathcal{H}}_i^R}^{\mathcal{H}_i^R} \oplus \bigoplus_{i \in [\tilde{d}] \setminus [r]} [1]_{\tilde{\mathcal{H}}_i^L}^{\mathcal{H}_i^L} \oplus [*]_{\tilde{\mathcal{H}}^\perp}^{\mathcal{H}^\perp},$$

$$e^{i\phi(2\Pi_R - I)} = \bigoplus_{i \in [k]} [e^{i\phi}]_{\mathcal{H}_i}^{\mathcal{H}_i} \oplus \bigoplus_{i \in [r] \setminus [k]} \begin{bmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{bmatrix}_{\mathcal{H}_i}^{\mathcal{H}_i} \oplus \bigoplus_{i \in [d] \setminus [r]} [e^{i\phi}]_{\mathcal{H}_i^R}^{\mathcal{H}_i^R} \oplus \bigoplus_{i \in [d] \setminus [r]} [e^{-i\phi}]_{\mathcal{H}_i^L}^{\mathcal{H}_i^L} \oplus [*]_{\mathcal{H}^\perp}^{\mathcal{H}^\perp},$$

¹Our notation maps top subspace to bottom: $\mathbf{r} \downarrow \mathbf{g}$

QSVT [GSLW19]: Let $\Phi = \{\phi_j\}_{j \in [n]} \in \mathbb{R}^n$; the QSVT protocol associated with Φ and block encoding U has circuit (taking n odd):

$$U_{\Phi} \equiv e^{i\phi_1(2\Pi_L - I)} U \prod_{j=1}^{(n-1)/2} e^{i\phi_{2j}(2\Pi_R - I)} U^{\dagger} e^{i\phi_{2j+1}(2\Pi_L - I)} U.$$

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$$\prod_{k=1}^n \left\{ \begin{array}{c} \text{---} \oplus \text{---} e^{i\phi_{2k}Z} \oplus \text{---} e^{i\phi_{2k-1}Z} \oplus \text{---} \\ \text{---} \boxed{U^\dagger} \boxed{\Pi_L} \text{---} \text{---} \boxed{\Pi_L} \boxed{U} \boxed{\Pi_R} \text{---} \text{---} \boxed{\Pi_R} \text{---} \end{array} \right\} = \begin{bmatrix} WP(\Sigma)V^\dagger & \dots \\ \vdots & \ddots \end{bmatrix}$$

Another approach:¹ *the cosine-sine decomposition*

¹The CSD is also useful in gate synthesis! [Tan25]

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It turns out one can produce *simultaneous* SVDs for multiple unitaries at once: $U_{ij} = V_i D_{ij} W_j^\dagger$ [TT23, PW94]

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Let $U \in \mathbb{C}^{d \times d}$ a unitary matrix partitioned into blocks of size $\{r_1, r_2\} \times \{c_1, c_2\}$:

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Then there exist unitaries $V_i \in \mathbb{C}^{r_i \times r_i}$ and $W_j \in \mathbb{C}^{c_j \times c_j}$ s.t.

$$\begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} = \begin{bmatrix} V_1 & \\ & V_2 \end{bmatrix} \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \begin{bmatrix} W_1 & \\ & W_2 \end{bmatrix}^\dagger,$$

where each D_{ij} is diagonal in $\mathbb{C}^{r_i \times c_j}$, possibly zero-padded.

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Specifically, we can write D as:

$$\begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} = \left[\begin{array}{c|cc} 0 & & I \\ & \textcolor{red}{C} & \textcolor{green}{S} \\ \hline I & & 0 \\ & \textcolor{green}{S} & \textcolor{red}{-C} \\ & 0 & -I \end{array} \right],$$

$$= \underbrace{\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}}_{\mathcal{X}_0 \rightarrow \mathcal{Y}_0} \oplus \underbrace{\begin{bmatrix} \textcolor{red}{C} & \textcolor{green}{S} \\ \textcolor{green}{S} & \textcolor{red}{-C} \end{bmatrix}}_{\mathcal{X}_C \rightarrow \mathcal{Y}_C} \oplus \underbrace{\begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}}_{\mathcal{X}_1 \rightarrow \mathcal{Y}_1}.$$

where $\textcolor{red}{C}$, $\textcolor{green}{S}$ are square w/ entries in $(0, 1)$, and $\textcolor{red}{C}^2 + \textcolor{green}{S}^2 = I$

Cosine-sine decomposition proof sketch

1. Start with the SVD of one block $U_{11} = V_1 D_{11} W_1^\dagger$
2. Compute QR decompositions of $U_{21} W_1$ and $U_{12}^\dagger V_1$, giving V_2, W_2 to make these operators upper-diagonal with nonnegative diagonal entries:

$$\begin{bmatrix} V_1 & \\ & V_2 \end{bmatrix}^\dagger \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix} \begin{bmatrix} W_1 & \\ & W_2 \end{bmatrix} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & V_2^\dagger U_{22} W_2 \end{bmatrix}.$$

3. Observing the rest of the unitary (whose rows and columns must be orthonormal), this forces the entries of D_{12}, D_{21} to be diagonal and $C^2 + S^2 = I$
4. Choose $W_2 \mapsto W_2'$ to correct D_{22} (free up to unitary).



The limits of qubitization



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In general, invariant subspaces for two reflections are only **one- and two-dimensional** [Jor75]

No good notion for *qu***d**itization



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Weaker variants of G-SVD do exist, applying to many matrices at once [PW94]



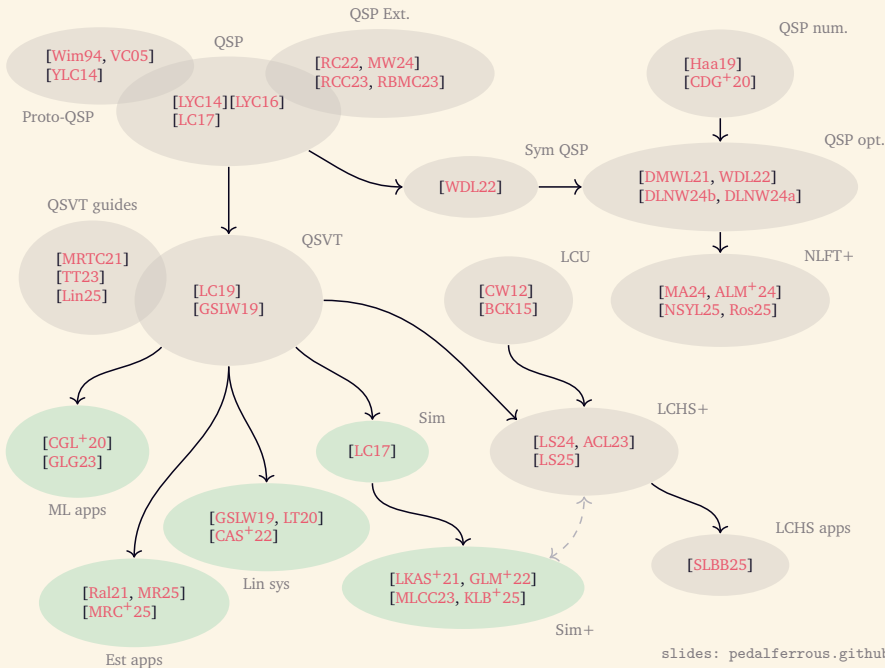
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Question: Weaker qubitization? Can we transform Jordan blocks [LS24] ? Irrep labels? Oracle-marked subspaces?



QSP is all* you need¹

¹[MRTC21]

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Matrix functions for large* linear operators

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Matrix functions for large* linear operators

$$\textcolor{red}{A} = \sum_k \xi_k |\tilde{\psi}_k\rangle\langle\psi_k| \xrightarrow[\text{QSVT}]{} \sum_k \textcolor{blue}{P}(\xi_k) |\tilde{\psi}_k\rangle\langle\psi_k| = \textcolor{blue}{P}(\textcolor{red}{A})$$

¹[[MRTC21](#)]

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$$A = \sum_k \xi_k |\tilde{\psi}_k\rangle\langle\psi_k| \xrightarrow{\text{QSVT}} \sum_k P(\xi_k) |\tilde{\psi}_k\rangle\langle\psi_k| = P(A)$$

Search:

Input **Grover oracle**, **apply** constant function

Low energy projection:

Input **Hamiltonian**, **apply** bandpass function

Inversion:

Input **sparse linear sys**, **apply** $1/x$ approximation

Simulation:

Input **Hamiltonian**, **apply** trig function ...

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 *Changing the polynomial changes the algorithm* 

QSVT in practice

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Linear system solving

Estimation tasks

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¹E.g., cryptography [[LMS21](#)], modified QPE [[Ral21](#)].

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⚠️ QSVT captures BQP-complete problems; if you ask for too much, it'll be expensive

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: Constant factors seem to matter [KREO25]; error propagation requires care [TT23, GSLW19]

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Beyond instantiation, diverse methods for (approximately) combining block encodings [LW19, VG25]

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👉 **A clean abstraction**; use of diverse functional analytic,
representation theoretic, and numerical linear algebraic
tools is made clear

A note on dequantization and QSVT

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Moral: If matrix is low rank, block encoding is QRAM-based, or your required precision is constant, then sketching methods work [[CGL⁺20](#), [GLG23](#), [EHG25](#)]

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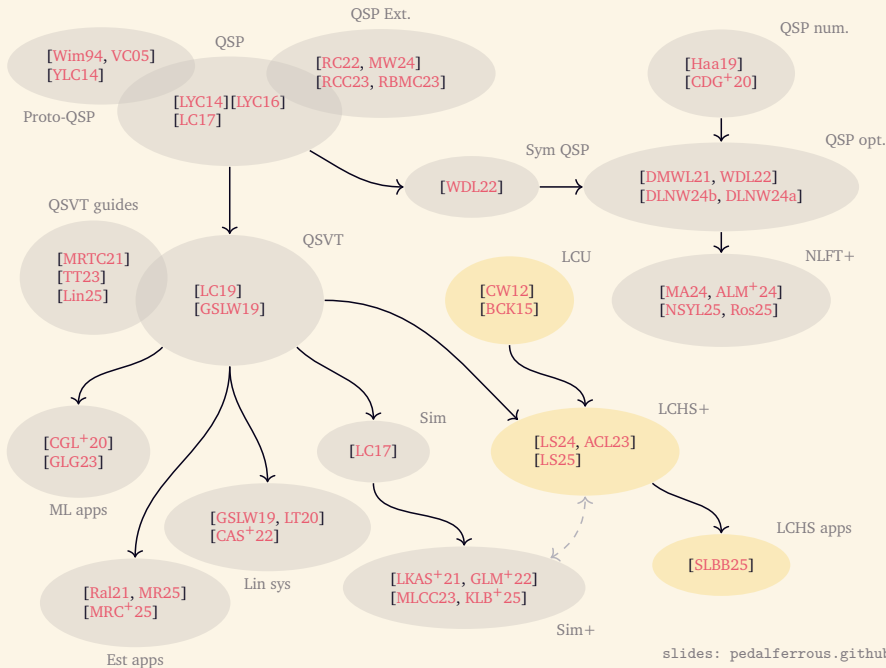
(Over)sampling and query access: analogues to quantum state preparation assumptions

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(Over)sampling and query access: analogues to quantum state preparation assumptions

In practice, there may be large polynomial separations even in these cases, but improvement is rapid!





The great family of block encoding algorithms

QSP [LC17], QSVT [GSLW19], QEP [LS24], G-QSP [MW24], M-QSP [RC22], LCU [CW12], LCHS [ACL23], randomized, parallelized, and hybrid variants!



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Comparison not always possible, and choices between these can depend on architecture and instance size!

Linear combination of unitaries (LCU) [CW12, BCK15]

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Let $V = \sum_k \alpha_k U_k$ for $\alpha_k > 0$ and U_k unitary. Let W be any unitary that acts as

$$W|0\rangle = \frac{1}{\sqrt{\alpha}} \sum_k \sqrt{\alpha_k} |k\rangle, \quad \alpha \equiv \sum_k \alpha_k.$$

Then

$$W^\dagger \left[\sum_k |k\rangle\langle k| \otimes U_k \right] W \equiv W^\dagger U W$$

is an $(\alpha, *, 0)$ block encoding of V . Often W, U are referred to as the PREPARE and SELECT operations.

LCU, QSVT, and Hamiltonian simulation

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Time-independent Hamiltonian sim for d -term LCU

Method	aux qubits	query comp
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QSVT insight: quick approximation of entire functions

$$\cos(xt) = J_0(t) + 2 \sum_{k=1}^{\infty} (-1)^k J_{2k}(t) T_{2k}(x),$$

$$\sin(xt) = 2 \sum_{k=0}^{\infty} (-1)^k J_{2k+1}(t) T_{2k+1}(x),$$

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Approximating $u(t)$ relies on time-ordered integral:

$$u(t) = U_0(t)u_0 + \int_0^t U_s(t)b(s) ds,$$

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$$\forall x \geq 0, \quad e^{-x} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \hat{f}(k) e^{-ikx} dk$$
$$\implies \forall t \geq 0, \quad e^{-(L+iH)t} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \hat{f}(k) e^{-i(kL+H)t} dk.$$

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 Integral quadrature can be technical [ACL23, TW14]

LCHS: utility and complexity

$$u_r \alpha t \log(1/\epsilon), \quad u_r = \|u\|_{L^\infty} / |u(t)|$$

In addition to pole, $\hat{f}$ is chosen to have its inverse Fourier coefficients decay exponentially

The key insight of [LS25] is that this decay only needs occur on infinite strip, not the half plane

⚠ Integral quadrature can be technical [ACL23, TW14]

Beyond e^{-x} , approximations for $1/x$ permit linear matrix equation solvers [SLBB25], e.g., $AX - XB = C$

The quantum linear system problem: $Ax = b$

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⚠️ QSVT exhibits suboptimal condition num dependence

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<i>Algorithm</i>	<i>Query comp.</i>	<i>N.b.</i>
HHL [HHL09]	$O(\kappa^2/\epsilon)$	VTAA
LCU [CKS17]	$O(\kappa^2 \text{polylog}(\epsilon^{-1}))$	VTAA
QSVT [GSLW19]	$O(\kappa^2 \log(\epsilon^{-1}))$	
VTAA [Amb10]	$O(\kappa \text{polylog}(\kappa \epsilon^{-1}))$	Overhead
Adiabatic [CAS ⁺ 22]	$O(\kappa \text{polylog}(\epsilon^{-1}))$	
Eig. filter [LT20]	$O(\kappa \log(\epsilon^{-1}))$	Adiabatic

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Moral: block encoding A^{-1} requires subnormalizing by κ , meaning $O(\kappa)$ amp, and thus $O(\kappa^2)$ query comp.

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 The utility of $\log(1/\epsilon)$ error dependence outside subroutines is debatable

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Open avenues in the near-term



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Weaker access
models and
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More flexible
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Past examples include **LCHS** [LS25], **multiproduct formulas** [LKW19], and **SKTs for QSP** [Ros25]



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We need more *end-to-end analysis* for *realistic problem instances*! Most operations can be fuzzed!



Block encoding algorithms & machine learning



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Great platform for dequantization efforts [[CGL⁺20](#)]



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Plenty of work left in realizing matrix manipulations efficiently (e.g., inference step in transformers [[GYC⁺25](#)])



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Great platform for dequantization efforts [CGL⁺20]

Investigations into quantum analogues of universal approximators [SAG⁺20, DGG⁺20, Ros25]



As always, *caveat lector*

Plenty of work left in realizing matrix manipulations efficiently (e.g., inference step in transformers [GYC⁺25])



Take-home



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We have *limited tools beyond qubitization* for nonlinear transformations of subsystem dynamics



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QSP/QSVT are one offshoot of a growing family of quantum algorithms for **efficiently manipulating sub-blocks of unitary matrices**

These algorithms and their hybrids enable importation of *vast techniques from applied mathematics*

We're not yet sufficiently optimistic

Regimes in which **about these algorithms!** *completely outperforms* another are *complex and evolving*

We have *limited tools beyond qubitization* for nonlinear transformations of subsystem dynamics

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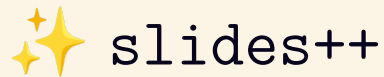
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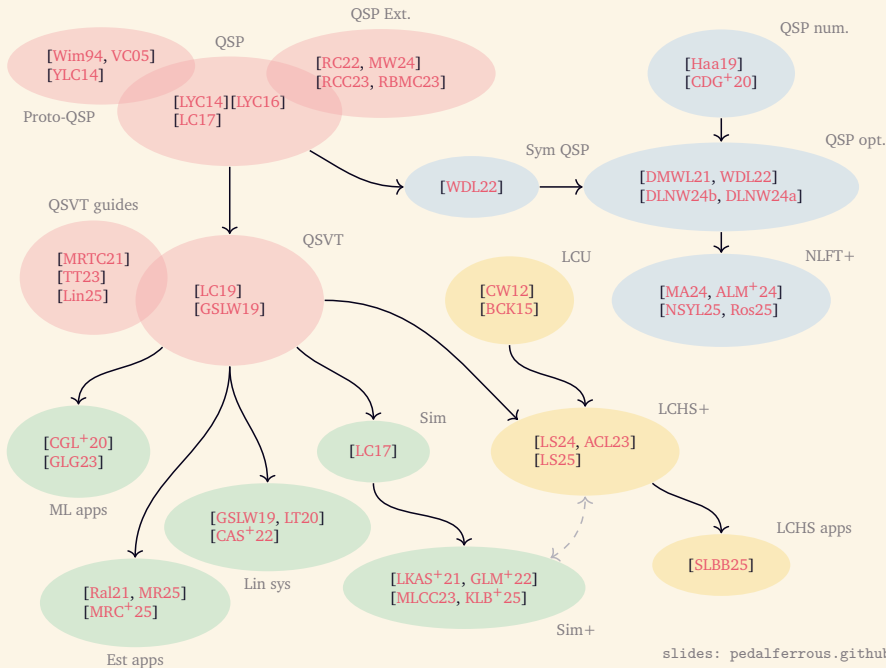
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Szegő functions

A real-valued measurable even function

$f : [0, 1] \rightarrow [-1, 1]$ is called Szegő if it satisfies

$$\int_0^1 \log |1 - f(x)^2| \frac{dx}{\sqrt{1 - x^2}} > -\infty \quad (9)$$

Szegő functions come with a norm showing they're a subset of square-summable

The work of [ALM⁺24] shows that Szegő functions satisfying $\|f\|_\infty \leq 1 - \eta$ admit unique QSP phases, and produce a unitary with a matrix element converging to f in a nice way, and that

$$\|\Phi - \Phi'\|_\infty \leq O(\eta^{-3}) \|f - f'\|_S. \quad (10)$$



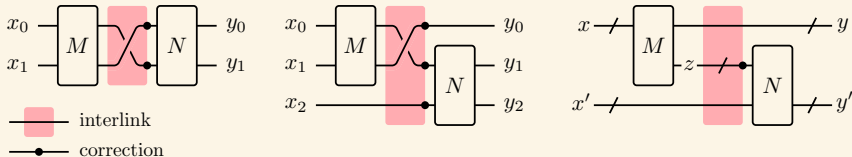
Hybrid methods in block encoding algorithms

Hybrid oscillator-qubit systems in experiment [[LSS⁺25](#)]

LCU of product formulas [[CST⁺21](#), [LKW19](#)]

Parallel, randomized, and self-composed QSP/QSVT
[[RCG23](#), [MRC⁺25](#), [MR25](#)]

Approximate or accelerated products for block encodings
[[LW19](#), [VG25](#)]



On the optimality of QSP/QSVT

Lower bound for eig. transformation; Thm. 73 [GSLW19]

Let $I \subseteq [-1, 1]$, $a \geq 1$ and suppose U is a $(1, a, 0)$ -block encoding of an unknown Hermitian matrix H with the promise that the spectrum of H lies within I . Let $f : I \rightarrow \mathbb{R}$, and suppose access to a quantum circuit V that implements a $(1, b, \varepsilon)$ -block encoding of $f(H)$ using T applications of U for all U satisfying the promise. Then for all $x \neq y \in I \cap [-1/2, 1/2]$ we have that

$$T = \Omega \left[\frac{|f(x) - f(y)| - 2\varepsilon}{|x - y|} \right]$$

Lower bound for quantum matrix functions; [MS24]

For any continuous function $f(x) : [-1, 1] \rightarrow [-1, 1]$, there is a 2-sparse Hermitian matrix A with $|A| \leq 1$ and two indices i, j such that $\Omega(\widetilde{\deg}_\varepsilon(f))$ queries to A are required in order to compute $\langle i | f(A) | j \rangle \pm \varepsilon/4$.